THREE REMARKS ON NORMS OF AFFINE COMPOSITION OPERATORS ON THE HARDY SPACE OF DIRICHLET SERIES

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Abstract. We answer two questions and a conjecture of Brevig and Perfekt on composition operators C_φ on the Hardy space $\mathscr{H}^2$ of Dirichlet series with affine symbols. For every admissible symbol $\varphi(s) = c + r2^{-s}$ with $r > 0$, the norm of C_φ is strictly larger than the supremum S^*_φ obtained by testing the adjoint on reproducing kernels of the minimal range. For $\varphi(s) = 1/2 + \xi(1 - 2^{-s})$, we prove that $\|C_\varphi\|^2 > 1/\xi$ for every $\xi > 0$, so that the conjectured value $\|C_\varphi\| = \xi^{-1/2}$ for small ξ fails. Finally, we give two affine symbols in three primes with the same mapping properties such that neither composition operator is subordinate to the other.

1. Introduction

Let $\mathscr{H}^2$ denote the Hilbert space of Dirichlet series $f(s) = \sum_{n\geq 1} a_n n^{-s}$ with $\|f\|^2_{\mathscr{H}^2} = \sum_n |a_n|^2 < \infty$. Its reproducing kernel is $K_w(s) = \zeta(s + \overline{w})$, $\operatorname{Re} w > 1/2$. Gordon and Hedenmalm [3] characterized the symbols φ for which the composition operator $C_\varphi f = f \circ \varphi$ is bounded on $\mathscr{H}^2$; among the Dirichlet series $\varphi = \sum_n c_n n^{-s}$, these form the class $\mathscr{G}$ of those which converge uniformly in $\operatorname{Re} s \geq \varepsilon$ for every $\varepsilon > 0$ and map $\operatorname{Re} s > 0$ into $\operatorname{Re} s > 1/2$. Brevig and Perfekt [1] studied the norms of such operators. Following them, an *affine symbol* is

$$\varphi_{\mathbf{c}}(s) = c + \sum_{j=1}^{d} c_j p_j^{-s},$$

where $p_1 < p_2 < \dots$ are the primes. It belongs to $\mathscr{G}$ exactly when $\operatorname{Re} c - 1/2 \geq r = \sum_j |c_j|$, and its *mapping properties* are $\omega = \varphi(+\infty) = c$ and $\Omega = \varphi(\operatorname{Re} s > 0) = \{|s - c| < r\}$, the latter by Kronecker's theorem [1, Lemma 3].

For the symbol $\varphi(s) = c + r2^{-s}$, the operator C_φ maps $\mathscr{H}^2$ into the closed subspace $\mathscr{H}^2_\Lambda$ spanned by $(2^j)^{-s}$, $j \geq 0$, whose reproducing kernel is

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$K_w^\Lambda(s) = (1 - 2^{-s-\overline{w}})^{-1}$. Brevig and Perfekt [1, Section 4] define

$$S_\varphi^* = \sup_{\operatorname{Re} w > 0} \frac{\|C_\varphi^* K_w^\Lambda\|_{\mathscr{H}^2}}{\|K_w^\Lambda\|_{\mathscr{H}^2}},$$

which is a lower bound for $\|C_\varphi\|$, and so is the analogous supremum S_φ obtained by testing C_φ on the normalized kernels of $\mathscr{H}^2$. Revising a question of Muthukumar, Ponnusamy and Queffélec [4], they ask the following [1, Section 4].

Question 1 (Brevig and Perfekt)**.** *If $\varphi(s) = c + r2^{-s}$ is in $\mathscr{G}$ and $r > 0$, does it hold that $\|C_\varphi\| = S_\varphi = S_\varphi^*$?*

They also prove that $(S_\varphi^*)^2 = 1/\xi$ when $\operatorname{Re} c - 1/2 = r = \xi \leq 1/4$ [1, Theorem 13], and remark that it is tempting to conjecture the following [1, Section 4].

Conjecture 1 (Brevig and Perfekt)**.** *For $\varphi(s) = 1/2 + \xi(1 - 2^{-s})$ and all sufficiently small $\xi > 0$, $\|C_\varphi\| = S_\varphi = S_\varphi^*$, and hence $\|C_\varphi\| = \xi^{-1/2}$.*

Finally, for two symbols φ and ψ with the same mapping properties, C_φ is *subordinate* to C_ψ if $\|C_\varphi f\|_{\mathscr{H}^2} \leq \|C_\psi f\|_{\mathscr{H}^2}$ for every $f \in \mathscr{H}^2$. Brevig and Perfekt prove that $C_{\varphi_{\mathbf{b}}}$ is subordinate to $C_{\varphi_{\mathbf{c}}}$ whenever $\mathbf{b}, \mathbf{c} \geq 0$ and $\mathbf{c}$ majorizes $\mathbf{b}$ [1, Theorem 5], and ask [1, Section 7.2] the following.

Question 2 (Brevig and Perfekt)**.** *If $\varphi_{\mathbf{b}}$ and $\varphi_{\mathbf{c}}$ are two affine symbols with the same mapping properties, is it true that either $C_{\varphi_{\mathbf{b}}}$ is subordinate to $C_{\varphi_{\mathbf{c}}}$, or $C_{\varphi_{\mathbf{c}}}$ is subordinate to $C_{\varphi_{\mathbf{b}}}$?*

By [1, Theorem 5], a counterexample requires at least three primes. The purpose of the present note is to answer both questions in the negative and to disprove the conjecture.

Theorem 1.

1. *If $\varphi(s) = c + r2^{-s}$ is in $\mathscr{G}$ and $r > 0$, then $S_\varphi^* < \|C_\varphi\|$.*
2. *If $\varphi(s) = 1/2 + \xi(1 - 2^{-s})$ with $\xi > 0$, then $\|C_\varphi\|^2 > 1/\xi$.*
3. *Let $\mathbf{a} = (9/20, 7/20, 1/5)$ and $\mathbf{b} = (49/100, 51/200, 51/200)$. For all sufficiently small $r > 0$ and $c = 1/2 + r$, the symbols $c + r\sum_{j=1}^3 a_j p_j^{-s}$ and $c + r\sum_{j=1}^3 b_j p_j^{-s}$ have the same mapping properties, and neither composition operator is subordinate to the other.*

Part (1) answers Question 1 in the negative for every admissible symbol. Part (2) disproves Conjecture 1, since $(S_\varphi^*)^2 = 1/\xi$ for $\xi \leq 1/4$, and part (3) answers Question 2.

2. The model operator

Let $\varphi(s) = c + r2^{-s}$ be in $\mathscr{G}$. The map $(2^j)^{-s} \mapsto z^j$ is unitary from $\mathscr{H}_\Lambda^2$ onto the Hardy space $H^2(\mathbb{D})$, and it carries C_φ to the operator $A\colon \mathscr{H}^2 \to H^2(\mathbb{D})$, $(Af)(z) = f(c + rz)$. Let $k_w(z) = (1 - \overline{w}z)^{-1}$ be the reproducing kernel of

$H^2(\mathbb{D})$. Since $\langle f, A^*k_w\rangle = (Af)(w) = f(c+rw)$, we have $A^*k_w = K_{c+rw}$, and hence the positive operator $T = AA^*$ on $H^2(\mathbb{D})$ satisfies

$$(Tk_w)(z) = K_{c+rw}(c+rz) = \zeta(2\operatorname{Re} c + rz + r\overline{w}). \tag{1}$$

We have $\|C_\varphi\|^2 = \|T\|$. The kernels of $\mathscr{H}^2_\Lambda$ correspond to k_w with $w = 2^{-v}$, $\operatorname{Re} v > 0$, that is, to $0 < |w| < 1$, so

$$\begin{aligned} (S^*_\varphi)^2 &= \sup_{0<|w|<1} F(w), \\ F(w) &= \frac{\langle Tk_w, k_w\rangle}{\|k_w\|^2} = (1-|w|^2)\,\zeta(2\operatorname{Re} c + 2r\operatorname{Re} w). \end{aligned} \tag{2}$$

Two reductions. For real τ the vertical translation $V_\tau f(s) = f(s+i\tau)$ is unitary on $\mathscr{H}^2$ and on $\mathscr{H}^2_\Lambda$, it maps the kernels of $\mathscr{H}^2_\Lambda$ to kernels of $\mathscr{H}^2_\Lambda$, and $f\circ(\varphi + i\tau) = (V_\tau f)\circ\varphi$ and $(f\circ\varphi)(\cdot + i\tau) = V_\tau(f\circ\varphi)$. Hence $\|C_\varphi\|$ and S^*_φ do not change if we replace c by $\operatorname{Re} c$, or replace 2^{-s} by $2^{-s-i\pi/\log 2} = -2^{-s}$. In particular we may change the sign of the coefficient of 2^{-s}, and (1) and (2) hold with r replaced by $-r$.

3. Proof of Theorem 1

Proof of part (1) when $\operatorname{Re} c - 1/2 > r$. Put $a = 2\operatorname{Re} c$. Since $a - 2r > 1$, the function F is continuous on the closed disc, positive inside and zero on the circle. So its supremum is attained at some w_0 with $|w_0| < 1$. Suppose that $\|T\| = F(w_0)$, and let $u = k_{w_0}/\|k_{w_0}\|$. Then $\langle(\|T\| - T)u, u\rangle = 0$, and since $\|T\| - T \geq 0$, we get $Tu = \|T\|u$. By (1) this means that

$$\zeta(a + r\overline{w_0} + rz) = \frac{\lambda}{1 - \overline{w_0}z}, \qquad z \in \mathbb{D},$$

for some $\lambda > 0$. Both sides are meromorphic in $\mathbb{C}$, so the identity holds in $\mathbb{C}$ away from the poles. As $z \to +\infty$ along the real axis the left-hand side tends to 1, while the right-hand side tends to 0 if $w_0 \neq 0$ and is constant if $w_0 = 0$; in the latter case the left-hand side is not constant. In either case we have a contradiction, so $\|T\| > F(w_0) = (S^*_\varphi)^2$. □

For part (2) we use a classical model of the kernel $(2 - z - \overline{w})^{-1}$. Let L_j be the Laguerre polynomials and $\ell_j(t) = \sqrt{2}e^{-t}L_j(2t)$, an orthonormal basis of $L^2(0,\infty)$, and let $U\colon H^2(\mathbb{D}) \to L^2(0,\infty)$ be the unitary map with $Uz^j = \ell_j$.

Lemma 1. *The operator B on $H^2(\mathbb{D})$ with $(Bk_w)(z) = (2 - z - \overline{w})^{-1}$ satisfies $UBU^{-1} = M$, where M is multiplication by e^{-2t}. In particular $\|B\| = 1$.*

Proof. The generating function $\sum_j L_j(x)y^j = (1-y)^{-1}\exp(-xy/(1-y))$ gives $(Uk_w)(t) = \sqrt{2}(1-\overline{w})^{-1}e^{-\beta t}$ with $\beta = (1+\overline{w})/(1-\overline{w})$. With $\beta' = (1+z)/(1-z)$ we get

$$\langle MUk_w, Uk_z\rangle = \frac{2}{(1-\overline{w})(1-z)}\int_0^\infty e^{-t(2+\beta+\beta')}\,\mathrm{d}t = \frac{1}{2 - z - \overline{w}},$$

since $2(1-\overline{w})(1-z)+(1+\overline{w})(1-z)+(1+z)(1-\overline{w}) = 4-2z-2\overline{w}$. This equals $\langle Bk_w, k_z\rangle$, and the kernels span a dense subspace. □

Proof of part (2). Here $c = 1/2+\xi$ and the coefficient of 2^{-s} is $-\xi$, so the argument of (1) applies with r replaced by $-\xi$, and $(Tk_w)(z) = \zeta(1+2\xi-\xi(z+\overline{w}))$. Write $\zeta(s) = (s-1)^{-1}+\psi(s)$, where ψ is entire and $\psi(1) = \gamma$ is Euler's constant. Then $T = \xi^{-1}B + R$, where $(Rk_w)(z) = \psi(1+2\xi-\xi(z+\overline{w}))$. The Taylor coefficients r_{jk} of this entire function of $(z,\overline{w})$ satisfy $\sum_{j,k}|r_{jk}| < \infty$, and $|\ell_j| \le \sqrt{2}$, so URU^{-1} is the integral operator with the continuous kernel $K(t,v) = \sum_{j,k} r_{jk}\ell_j(t)\ell_k(v)$. Since $\ell_j(0) = \sqrt{2}$, we have $K(0,0) = 2\psi(1) = 2\gamma > 0$, and we fix $\delta > 0$ with $K \ge \gamma$ on $[0,\delta]^2$.

For $0 < \varepsilon < \delta$ let $h = t^{-1}\mathbf{1}_{[\varepsilon,\delta]}$ and $L = \log(\delta/\varepsilon)$. Since $1-e^{-2t} \le 2t$, we have $\int_0^\infty (1-e^{-2t})h^2\,\mathrm{d}t \le 2L$, and since $h \ge 0$ and $K \ge \gamma$ on the support of $h\otimes h$, the remainder contributes at least $\gamma(\int h)^2 = \gamma L^2$. By Lemma 1, therefore,

$$\langle (UTU^{-1} - \xi^{-1})h, h\rangle \ge -2\xi^{-1}L + \gamma L^2,$$

which is positive when $L > 2/(\gamma\xi)$. Hence $\|C_\varphi\|^2 = \|T\| > \xi^{-1}$. For $\xi \le 1/4$, [1, Theorem 13] gives $(S_\varphi^*)^2 = \xi^{-1}$. □

Proof of part (1) when $\operatorname{Re} c - 1/2 = r$. By the reductions we may assume that $c = 1/2+r$, so that $F(w) = (1-|w|^2)\zeta(1+2r+2r\operatorname{Re} w)$. The function F is continuous on the open disc and tends to 0 at every boundary point except -1. Writing $w = -1+h$ with $\operatorname{Re} h > 0$, we have $1-|w|^2 \le 2\operatorname{Re} h$ and, by $\zeta(1+x) \le 1+1/x$, $F(w) \le 1/r + 2\operatorname{Re} h$. So $\limsup_{w\to -1} F(w) \le 1/r$. On the other hand, $(S_\varphi^*)^2 \ge 1/r$ [1, Theorem 13(a)]. If $\sup F > 1/r$, the supremum is attained at an interior point, and the argument for $\operatorname{Re} c - 1/2 > r$ applies verbatim. If $\sup F = 1/r$, then part (2), applied with $\xi = r$ after changing the sign of the coefficient, gives $\|C_\varphi\|^2 > 1/r = (S_\varphi^*)^2$. □

Proof of part (3). Both vectors are positive with sum 1, so both symbols belong to $\mathscr{G}$ with $\operatorname{Re} c - 1/2 = r = \sum_j ra_j = \sum_j rb_j$, and both have $\omega = c$ and $\Omega = \{|s-c| < r\}$. For $n \ge 2$ and the unit vector $f_n(s) = n^{-s}$, put $x = r\log n$. Then

$$C_{\varphi_{r\mathbf{a}}} f_n(s) = n^{-c}\prod_{j=1}^{3}\exp(-xa_jp_j^{-s}) = n^{-c}\prod_{j=1}^{3}\sum_{k\ge 0}\frac{(-xa_j)^k}{k!}(p_j^k)^{-s},$$

and since the products $p_1^{k_1}p_2^{k_2}p_3^{k_3}$ are distinct,

$$\|C_{\varphi_{r\mathbf{a}}} f_n\|^2 = n^{-2c}\prod_{j=1}^{3} I_0(2xa_j), \qquad I_0(2y) = \sum_{k\ge 0}\frac{y^{2k}}{(k!)^2},$$

and similarly for $\mathbf{b}$. Let $\rho(x) = \prod_j I_0(2xa_j)/I_0(2xb_j)$. Since $\log I_0(2y) = y^2 + O(y^4)$ and $\sum_j a_j^2 - \sum_j b_j^2 = -103/20000$, we have $\rho(x) < 1$ for all sufficiently

small $x > 0$. Since $I_0(2y) \sim e^{2y}/\sqrt{4\pi y}$ as $y \to \infty$ and $\sum_j a_j = \sum_j b_j$,

$$\lim_{x\to\infty} \rho(x) = \Big(\frac{b_1 b_2 b_3}{a_1 a_2 a_3}\Big)^{1/2} = \Big(\frac{2023}{2000}\Big)^{1/2} > 1.$$

Fix r so small that $\rho(r\log 2) < 1$, and then n so large that $\rho(r\log n) > 1$. Then $\|C_{\varphi_{r\mathbf{a}}} f_2\| < \|C_{\varphi_{r\mathbf{b}}} f_2\|$, so $C_{\varphi_{r\mathbf{b}}}$ is not subordinate to $C_{\varphi_{r\mathbf{a}}}$, and f_n gives the reverse strict inequality. Hence neither operator is subordinate to the other. □

4. Concluding remarks

The size of the excess. Part (2) determines only the sign of $\|C_\varphi\|^2 - \xi^{-1}$, with an exponentially small lower bound as $\xi \to 0$. Fang, Guo, Mishra and Muthukumar [2] obtain $\|C_\varphi\|^2 = \xi^{-1} + O(1)$; their estimate does not decide the sign. It would be interesting to determine the limit of $\|C_\varphi\|^2 - \xi^{-1}$ as $\xi \to 0$, if it exists.

Incomparable vectors. The vectors in part (3) are incomparable in the majorization order, as they must be by [1, Theorem 5]: the partial sums of the decreasing rearrangements are $(0.45, 0.80)$ and $(0.49, 0.745)$. The logarithm of $\|C_\varphi f_n\|^2$ is a power series in x^2 whose coefficients are fixed multiples of the moments $\sum_j a_j^{2k}$, and the example exploits that a vector with smaller second moment can have smaller product. The condition on r in part (3) is mild: numerically $\rho(x) < 1$ exactly for $0 < x < 2.679\ldots$, and for $r = 1$, $c = 3/2$ one finds $\rho(\log 2) \approx 0.99797$ and $\rho(\log 100) \approx 1.00912$. It would be interesting to characterize the pairs $\mathbf{b}, \mathbf{c}$ for which subordination holds.

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References

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